

Condensate-quant

Code collection for detecting, segmenting, and quantifying biomolecular condensates in live-cell super-resolution microscopy images. The repository includes workflows for multi-channel chromatic alignment, Noise2Void self-supervised denoising, Richardson-Lucy deconvolution, HMRF-based condensate segmentation, foci-to-condensate spatial analysis, and downstream visualization.

The scripts in this repository are associated with the paper **Coordinated dynamics of condensates and enhancer-promoter looping revealed by live-cell imaging**.

Repository Structure

```
Condensate-quant/
  README.md
  Function/                # MATLAB utility functions
  bfmtlab/                 # Bio-Formats MATLAB tools
  CondensateAnalysis/     # Condensate segmentation and analysis
  functions
  plotFunction/           # Plotting utilities
  onnx/                   # Pre-trained Noise2Void ONNX models
  PSF/                    # PSF files for different channels and
  imaging modes
  Step0_channel_alignment/ # Multi-channel chromatic alignment
  Step1_pytorch-noise2void/ # PyTorch Noise2Void training and
  testing
  Step2_deconvolution-segmentation/ # Denoising, deconvolution,
  segmentation, and quantification
  Step3_downstream_analysis/ # Experiment-specific downstream spatial
  analysis
```

Workflow Overview

The analysis workflow contains four main steps:

1. `Step0_channel_alignment/`
 - Estimate geometric transformation matrices between fluorescence channels using bead calibration data.
 - Apply chromatic correction to ND2 or batch image datasets through a MATLAB App.
2. `Step1_pytorch-noise2void/`
 - Train a Noise2Void-style self-supervised denoising model using PyTorch.
 - Save PyTorch checkpoints and export ONNX models for MATLAB inference.
3. `Step2_deconvolution-segmentation/`
 - Read OME-TIFF time-series image stacks.
 - Apply sliding-window denoising using a pre-trained ONNX Noise2Void model.
 - Perform Richardson-Lucy deconvolution using experimentally measured PSFs.
 - Generate nucleus masks and identify condensate regions using HMRF segmentation.
 - Export condensate masks, centers, boundaries, interfaces, and time-resolved morphometric measurements.

4. `Step3_downstream_analysis/`

- Run experiment-specific downstream analysis for RNA, DNA, enhancer/promoter, and condensate imaging assays.
- Detect DNA/RNA foci, extract local ROIs around foci, segment OCT4/BRD4 condensates, and calculate distances from foci to condensate boundary, interface, centroid, and center.
- Generate summary matrices and visualization panels including scatter plots and density contours.

Requirements

MATLAB

A recent MATLAB version is recommended, together with the following toolboxes:

- Image Processing Toolbox
- Deep Learning Toolbox
- Statistics and Machine Learning Toolbox
- Computer Vision Toolbox, for some registration and geometric transformation functions

Bio-Formats MATLAB tools are included in:

```
Function/bfmatlab/
```

Before running MATLAB scripts, add the repository and all subfolders to the MATLAB path:

```
addpath(genpath('C:\path\to\Condensate-quant'));
```

Python

Noise2Void training and testing scripts are located in `Step1_pytorch-noise2void/`. A conda environment is recommended:

```
cd Step1_pytorch-noise2void
conda create -n n2v python=3.10 -y
conda activate n2v
pip install -r requirements.txt
```

On Windows, set the following option in both `train_pipeline.py` and `test_pipeline.py`:

```
num_workers = 0
```

This avoids common multiprocessing issues during data loading.

Data

Example data, raw data information, and organized downstream data templates are listed under the corresponding analysis folders, for example:

```
Step2_deconvolution-segmentation/DataSource.txt
Step3_downstream_analysis/3color-RNA-2Condensate/DataSource.txt
Step3_downstream_analysis/4color-DNA-RNA-2Condensate/*/DataSource.txt
Step3_downstream_analysis/4color-Promoter-RNA-Enhancer-
Condensate/*/DataSource.txt
Step3_downstream_analysis/3color-Promoter_RNA_SCR/
```

The current data source points to the Zenodo DOI:

```
10.5281/zenodo.18548898
```

`Step2_deconvolution-segmentation/` and `Step3_downstream_analysis/3color-RNA-2Condensate/` contain example inputs and generated outputs, which can be used to inspect expected file naming and output organization.

Step 0: Multi-Channel Chromatic Alignment

Directory:

```
Step0_channel_alignment/
```

Main files:

- `script_get_transformation_matrix.m`: computes channel-to-channel transformation matrices from fluorescent bead localization results.
- `ImageRegistrator.mlapp`: MATLAB App for applying transformation matrices to biological image data.
- `tform_20260123.mat`: example or precomputed transformation matrix file.

Input

The registration script expects bead localization CSV files exported from Fiji TrackMate. Each calibration dataset should usually contain:

```
*_405channel.csv
*_488channel.csv
*_560channel.csv
*_640channel.csv
```

Usage

1. Modify `filename_list` in `script_get_transformation_matrix.m` to point to your bead calibration datasets.
2. Run the script to generate 2D and 3D geometric transformation matrices.
3. Open `ImageRegistrator.mlapp` and select the input images, output directory, and transformation matrix file.
4. Choose the imaging mode, either Live-SR or Confocal, and export chromatically corrected images.

Step 1: Noise2Void Denoising Model Training and Testing

Directory:

```
step1_pytorch-noise2void/
```

Main files:

- `train_pipeline.py`: trains the Noise2Void model.
- `test_pipeline.py`: runs denoising with a trained model.
- `noise2void/`: core model, data processing, training, and inference code.
- `pth/`: PyTorch checkpoint output directory.
- `onnx/`: ONNX model output directory.
- `requirements.txt`: Python dependencies.

Training

Modify the dataset path and training parameters in `train_pipeline.py`:

```
datasets_path = 'your/training/tif/folder'
GPU = '0'
patch_xy = 128
patch_t = 8
pth_dir = './pth'
```

Then run:

```
cd step1_pytorch-noise2void
python train_pipeline.py
```

Training outputs are saved in `pth/`, and exported ONNX models are saved in `onnx/`.

Testing

Modify the dataset path, model folder name, and output directory in `test_pipeline.py`:

```
datasets_path = 'your/input/tif/folder'
denoise_model = 'your_model_folder_name'
output_dir = 'your/output/folder'
```

Then run:

```
python test_pipeline.py
```

Step 2: Denoising, Deconvolution, Condensate Segmentation, and Quantification

Directory:

```
step2_deconvolution-segmentation/
```

Main script:

```
denoise_deconv_condensate_2D_time_series_HMRF_boundary.m
```

Main Inputs

- OME-TIFF time-series images (`.tif`)
- Noise2Void ONNX model, for example:

```
onnx/N2V_3D_OCT4_BRD4_liveSR_125_8_E10_xy3z0.onnx
```

- Experimentally measured PSF, for example:

```
PSF/psf_SR_channel561_2D.tif
```

- Key parameters:
 - `pixelSize`: physical pixel size in nm.
 - `resize_factor`: sub-pixel interpolation factor.
 - `min_radius`: minimum condensate radius threshold in nm.
 - `min_partition_coefficient`: minimum enrichment threshold.
 - `nclust` and `seg_point`: HMRF cluster number and condensate class threshold.

Parameters to Modify Before Running

Open `denoise_deconv_condensate_2D_time_series_HMRF_boundary.m` and update the following paths and parameters for your data:

```
onnx_path = '../onnx/N2V_3D_OCT4_BRD4_liveSR_125_8_E10_xy3z0.onnx';
psf_SR_560 = TIFFreader('../PSF/psf_SR_channel561_2D.tif', 'double');
filepath_list = {'path/to/your/tif/folder/'};
pixelSize = 95;
resize_factor = 10;
min_radius = 50;
min_partition_coefficient = 1.2;
```

Then run the script in MATLAB.

Outputs

For each input `.tif` file, the script creates an output folder with the same base name and generates:

```
*-denoised.tif           # Noise2Void denoising result
*-denoised-deconv.tif    # Denoised and deconvolved result
*-CDmask.tif             # Condensate mask with boundary visualization
*-CDcenter.tif           # Condensate centers
*-CDinterface.tif        # Condensate interfaces
*-CDboundary.tif         # Condensate boundaries
*-HMRFseg.png            # HMRF segmentation preview for the first frame
*.mat                   # Image data, masks, condensate structures, and
quantification results
```

The `condensate_result` structure contains:

```
CDcenter  
CDinterface  
CDboundary  
CDmask  
Labels  
area
```

The second half of the main MATLAB script includes statistical analysis and visualization of condensate number, area, and radius over time. If preprocessing results are already available as `.mat` files, the statistics section can be run independently without repeating denoising, deconvolution, and segmentation.

Step 3: Downstream Analysis

Directory:

```
step3_downstream_analysis/
```

Step 3 contains experiment-specific MATLAB scripts for measuring spatial relationships between DNA/RNA foci and OCT4/BRD4 condensates. These scripts reuse the denoising, deconvolution, HMRF segmentation, ROI extraction, and distance-measurement functions from `Function/`.

Subfolders

```
step3_downstream_analysis/  
  3color-RNA-2Condensate/  
  4color-DNA-RNA-2Condensate/  
  4color-Promoter-RNA-Enhancer-Condensate/  
  3color-Promoter_RNA_SCR/
```

3-Color RNA and Two-Condensate Analysis

Directory:

```
step3_downstream_analysis/3color-RNA-2Condensate/
```

Main script:

```
denoise_deconv_RNA_condensate_2D_HMRF_boundary.m
```

This workflow analyzes 3-color images containing RNA, OCT4, and BRD4 channels. It detects RNA foci, extracts local ROIs around RNA sites, segments OCT4 and BRD4 condensates using HMRF, and calculates distances between RNA foci and condensate boundaries and centroids.

Expected channel labels:

```
RNA, OCT4, BRD4
```

4-Color DNA/RNA and Two-Condensate Analysis

Directory:

```
step3_downstream_analysis/4color-DNA-RNA-2Condensate/
```

Main script:

```
denoise_deconv_foci_condensate_average_HMRF_boundary.m
```

This workflow analyzes 4-color images containing DNA, RNA, OCT4, and BRD4 channels. It supports separate cell lists for RNA-positive and RNA-negative cells, detects DNA/RNA foci, segments OCT4 and BRD4 condensates, calculates foci-to-condensate spatial metrics, and generates scatter and contour visualizations.

Expected subfolders:

```
Promoter_RNA_OCT4_BRD4/  
SCR_RNA_OCT4_BRD4/
```

Each subfolder should contain:

```
cell_with_RNA.txt  
cell_without_RNA.txt  
Imaging_data.tif
```

4-Color Promoter/RNA/Enhancer and Single-Condensate Analysis

Directory:

```
step3_downstream_analysis/4color-Promoter-RNA-Enhancer-Condensate/
```

Main script:

```
denoise_deconv_PRE_condensate_averaged_HMRF_boundary.m
```

This workflow analyzes 4-color images containing promoter, RNA, enhancer/SCR, and one condensate channel. It extracts promoter/RNA/enhancer foci, segments the selected OCT4 or BRD4 condensate channel, calculates distances from each foci type to condensate boundaries and centroids, and summarizes promoter-RNA-enhancer geometry.

Expected subfolders:

```
Promoter_RNA_SCR_BRD4/  
Promoter_RNA_SCR_OCT4/
```

Each subfolder should contain:

```
Cell_with_RNA.txt
Cell_without_RNA.txt
Imaging_data.tif
```

3-Color Promoter/RNA/Enhancer Distance Analysis

Directory:

```
step3_downstream_analysis/3color-Promoter_RNA_SCR/
```

Main script:

```
denoise_deconv_PRE_distance.m
```

This workflow analyzes 3-color LiveSR images containing promoter, RNA, and enhancer/SCR channels. It denoises and deconvolves the image stacks, detects promoter/RNA/enhancer foci, extracts local ROIs around foci, and calculates promoter-enhancer distances for RNA-positive and RNA-negative cells.

Expected channel labels:

```
Promoter, RNA, Enhancer
```

Each subfolder should contain:

```
Cell_with_RNA.txt
Cell_without_RNA.txt
Imaging_data.tif
```

`Cell_with_RNA.txt` and `Cell_without_RNA.txt` should list the organized ON/OFF raw image filenames. `zstep_sheet.csv` should contain one row per processed `.mat` file, with at least the columns `name` and `var2`, where `var2` is the z-step size in nm used for 3D distance calculation.

Step 3 Outputs

Depending on the workflow, outputs may include:

```
*-denoised.ome.tif
*-denoised-deconv.ome.tif
*-RNA.tif
*-DNA.tif
*-Promoter.tif
*-Enhancer.tif
*-Condensate.tif
*-Center-roi.tif
*-roi.tif
*-roi-bicubic.tif
*-HMRfseg.png
*-CDmask.tif
*-CDcenter.tif
*-CDinterface.tif
*-CDboundary.tif
```



```
*.mat  
*.png  
cell_analysis_errors.txt
```

The output `.mat` files typically store:

```
foci_result  
condensate_result  
img_series_max  
nucleus_mask  
roi_window  
channel_labels  
pixelSize  
resize_factor  
dist_summary_withRNA  
dist_summary_woRNA  
epdist_rna_on  
epdist_rna_off
```

Parameters to Check Before Running

Before running Step 3 scripts, verify:

- The script is launched from its own Step 3 subfolder, or relative paths are updated accordingly.
- `onnx_path` points to the correct Noise2Void ONNX model.
- PSF paths match the imaging mode and channel order.
- `filepath_list` points to the intended experiment folders.
- `Cell_with_RNA.txt` and `Cell_without_RNA.txt` contain valid image filenames.
- `channel_labels` match the actual image channel order.
- `pixelSize`, `resize_factor`, `roi_width`, `nclust`, and `seg_point` are appropriate for the dataset.
- `min_radius` and `min_partition_coefficient` are appropriate for condensate filtering. The current Step 3 defaults are `min_radius = 50` nm and `min_partition_coefficient = 1.2`.

Notes

- Several scripts contain example relative or absolute paths. Update them to match your local data paths before running.
- Step 3 batch-processing scripts write `cell_analysis_errors.txt` in each output directory when an individual cell fails; failed cells are skipped so the remaining cells can continue processing.
- Step 3 scripts are experiment-specific and are best treated as analysis templates. Check channel order, z-layer selection logic, and foci-matching thresholds before applying them to a new dataset.
- On Windows, set `num_workers = 0` in the Python scripts.
- If GPU inference is unavailable in MATLAB, set:

```
is_GPU_avaliable = false;
```

- Choose a PSF file that matches both the imaging mode and fluorescence channel, such as `SR/noSR` and `405/488/561/642`.
- HMRF segmentation can be sensitive to `nclust`, `seg_point`, `min_radius`, and `min_partition_coefficient`. It is recommended to inspect the output images on a small subset of data before batch processing.

Citation

If you use this code, please cite the corresponding paper:

Coordinated dynamics of condensates and enhancer-promoter looping revealed by live-cell imaging

For questions, contact:

wangbo@stu.pku.edu.cn